

Multi-Agent Fraud Detection Network

Real-Time Fraud Prevention System

Team: Alverse

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Meet the Team – Alverse



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The Scale of the Problem

\$40B+

Global fraud losses annually

300%

Rise in UPI/digital fraud

72hrs

Avg detection delay

23%

False positive rate in legacy systems

Surge in Digital Fraud

- UPI, card & crypto fraud increasing rapidly
- Sophisticated attack patterns emerging daily

Static Rule-Based Systems

- Cannot adapt to new fraud patterns
- High false positive rates disrupt operations

Delayed Detection

- Legacy systems detect fraud after 24–72 hours
- Significant losses occur before intervention

No Relationship Analysis

- Isolated transactions only — no network view
- Coordinated fraud rings go undetected



Real-Time Detection

Detect and block fraudulent transactions within milliseconds using event-driven streaming



Multi-Layer Intelligence

Deploy 6 independent agents in parallel — behavioral, graph, statistical, and rule-based



Reduce Fraud Losses

Target 95%+ fraud capture rate, reducing financial losses across UPI, card, and crypto



Improve Accuracy

Minimize false positives through multi-agent consensus and confidence-weighted scoring



Graph-Based Relationship Analysis

Uncover fraud rings and money mule networks via Neo4j graph intelligence



Scalable & Compliant Design

Cloud-native, containerized architecture ready for enterprise-scale deployment

System Overview



Multi-Agent System



Event-Driven Architecture



Kafka-Based Streaming



Parallel Processing

Distributed Agent Network:

Six specialized AI agents operate independently and in parallel, each focusing on a distinct fraud signal domain.

Kafka Event Streaming:

Apache Kafka enables low-latency, high-throughput transaction streaming across all agent topics (txn.raw, txn.scored, txn.alert, txn.response).

Consensus-Based Decision Making:

Risk scores from all agents are aggregated by the Risk Scorer; the Decision Engine applies a final verdict using weighted confidence thresholds.

Cloud-Native & Containerized:

Docker + Kubernetes deployment ensures horizontal scalability, zero-downtime rollouts, and multi-cloud portability.

Architecture Diagram



6 Independent Kafka Topics & Agents:

txn.raw → Transaction Monitor

txn.raw → Behaviour Analyzer

txn.raw → Graph Fraud Detector

txn.scored → Risk Scorer

txn.alert → Decision Engine

txn.response → Compliance Logger

Agents Overview

1

Transaction Monitor



Real-time ingestion & rule-based validation of incoming transactions

2

Behaviour Analyzer



ML-based profiling of user spending patterns; detects deviations using Z-score

3

Graph Fraud Detector



Neo4j-powered graph traversal to identify fraud rings and shared account networks

4

Risk Scorer



Aggregates confidence scores from all agents using weighted ensemble method

5

Decision Engine



Applies threshold logic: ALLOW / REVIEW / BLOCK based on aggregated risk score

6

Compliance Logger



Immutable audit trail of every transaction decision for regulatory reporting

Key Technologies

Event Streaming

Apache Kafka

High-throughput, fault-tolerant message broker for real-time transaction streaming across all agents

Graph Database

Neo4j

Graph-native database for storing transaction relationships and detecting fraud ring patterns

ML Classification

XGBoost

Gradient boosting classifier for high-accuracy supervised fraud classification on labeled datasets

Anomaly Detection

Isolation Forest

Unsupervised ML algorithm to identify outlier transactions without labeled fraud data

Backend
Framework

FastAPI

High-performance async Python API framework powering the agent orchestration layer and dashboard UI

Containerization &
Orch.

Docker + K8s

Docker containers for each agent; Kubernetes orchestration for auto-scaling and rolling deployments

01



Transaction Received

Payment event ingested via REST API; published to Kafka topic txn.raw with metadata: amount, device, location, beneficiary

02



Parallel Agent Execution

Transaction Monitor, Behaviour Analyzer, and Graph Fraud Detector consume txn.raw simultaneously — zero sequential bottleneck

03



Risk Aggregation

Each agent publishes its confidence score to txn.scored; Risk Scorer consumes all scores and computes a weighted ensemble risk value

04



Decision Generation

Decision Engine evaluates risk score against thresholds: <0.3 ALLOW | 0.3–0.7 REVIEW | >0.7 BLOCK — publishes to txn.alert

05



Logging & Compliance

Compliance Logger writes immutable audit record to txn.response — full history of decision, scores, agent votes, and timestamps

Scenario: High-Risk Account Takeover Attempt — Foreign Login + New Beneficiary Transfer

Transaction Monitor

Transaction 4.2× above user average

Score
0.61

Behaviour Analyzer

Login from IP in Germany — user normally in India

Score
0.79

Graph Fraud Detector

Beneficiary linked to 3 flagged fraud accounts

Score
0.91

Final Result

Risk Score: 0.87 / 1.0

Threshold: > 0.70

Decision:  BLOCK

Response Time: < 180ms

12,847

Transactions Today

99.2%

Detection Accuracy

2.1%

False Positive Rate

147ms

Avg Response Time

Real-Time Fraud Alerts

Txn ID	Type	Risk	Status
TXN-9921	Account Takeover	0.91	BLOCKED
TXN-9918	Duplicate Payment	0.74	BLOCKED
TXN-9914	Velocity Abuse	0.58	REVIEW
TXN-9908	New Beneficiary	0.42	REVIEW
TXN-9901	Normal Payment	0.11	ALLOWED

FastAPI UI

High-performance async dashboard with real-time WebSocket updates

Fraud Analytics

Interactive charts: fraud trends, agent performance, top fraud types

Compliance Reports

Exportable PDF/CSV audit logs with full decision lineage

System Monitoring

Kafka consumer lag, agent health, and latency metrics via Grafana



Real-Time Detection

Sub-200ms decision latency via Kafka streaming and parallel agent execution



Multi-Agent Architecture

Six specialized agents provide comprehensive coverage; each independently scalable



Graph Intelligence

Neo4j-powered relationship analysis catches coordinated fraud rings invisible to rule engines



Scalable by Design

Docker + Kafka partitioning enables horizontal scaling to handle millions of daily transactions



95%+ Accuracy

Multi-agent consensus with ensemble ML minimizes false positives and missed fraud



Regulatory Compliance Ready

Immutable audit trails and GDPR-aligned data governance built into the core architecture

Phase 2

AI Explainability (XAI)

SHAP/LIME integration to provide human-readable explanations for every fraud decision to regulators and end-users

Phase 2

Blockchain Audit Trail

Immutable, tamper-proof transaction records using distributed ledger technology for enhanced regulatory compliance

Phase 3

Advanced Graph ML

Graph Neural Networks (GNN) for deeper pattern recognition across millions of interconnected transaction nodes

Phase 3

Cloud Auto-Scaling

AWS/GCP native deployment with serverless Kafka (MSK/Pub-Sub) and managed Kubernetes for elastic scaling

Phase 4

ERP & Banking Integration

Direct API connectors to SAP, Oracle Financials, and core banking systems for seamless enterprise adoption

Phase 4

Federated Learning

Cross-institution fraud model training without sharing raw data — privacy-preserving collaborative intelligence

High-Accuracy Fraud Detection For The Modern Digital Economy

Scalable. Accurate. Industry-Ready.



Multi-Agent Architecture

1

Six specialized agents provide comprehensive, resilient fraud coverage unmatched by single-model approaches

Real-Time Performance

2

Sub-200ms detection latency with 99.2% uptime SLA — production-grade reliability

Industry-Ready Stack

3

Kafka, Neo4j, Docker, FastAPI — battle-tested open standards with enterprise community support

Regulatory Compliance Built-In

4

Audit trails, GDPR alignment, and GAAP-equivalent transaction logging from day one

Thank You

Multi-Agent Fraud Detection Network

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Powered by Kafka · Neo4j · XGBoost · FastAPI · Docker

